Text Retrieval and Search Engines

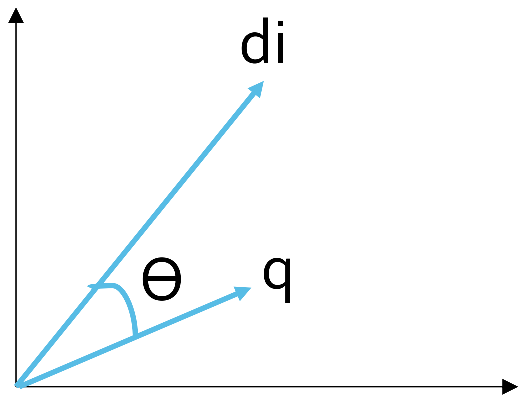
Lecture 8 - Probabilistic Retrieval Model

Some slides adapted from the companion website to the "Introduction to IR" book by Manning, Raghavan and Schutz & Victor Lavrenko

Semester 1 – 2026

Why Using Vector Space?

- Why does it work? no notion of relevance anywhere
- A well-formed algebraic space for retrieval
  - Key Idea: user's query can be views as a (very) short document
- Query becomes a vector in the same space as the documents
- Can measure each document's proximity to it



Why Do We Need a Probabilistic Model?

- Motivation
  - Speech recognition
    - Given that we see “John” and “feels”, which word is more likely to appear next? “happy” or “habit”
  - Classification
    - Given that we observe “football” three times and “game” once in a news article, how likely is it about “sports”?
  - Ad hoc retrieval
    - Given that a user is interested in sports news, how likely would the user use “football” in a query?
- Mathematical formalism for relevant / non-relevant sets

The Document Ranking Problem

- We have a collection of documents
- User issues a query
- A list of documents needs to be returned
- Ranking method is core of an IR system:
  - In what order do we present documents to the user?
    - We want the “best” document to be first, best second to be second, …
  - Idea: Rank by probability of relevance of the document w.r.t. information need
    - P(relevant|document,query)

Probability Ranking Principle (PRP)

- Robertson (1977)

  "If a reference retrieval system's response to each request is a **ranking** of the documents in the collection in order of decreasing **probability of relevance** to the user who submitted the request,

  Where the **probabilities** are **estimated** as **accurately** as possible on the basis of whatever data have been made available to the system for this purpose,

  The overall **effectiveness** of the system to its user **will be** the **best** that is obtainable on the basis of those data."

- Basis for most probabilistic approaches to IR

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- Based on whatever data is available to system
  - $P_{est}(relevant|document, session, context, collection, user\ profile, \ldots)$
- Best possible estimate one can achieve with that data
  - Recipe for a perfect IR system: just need $P_{est}(R|\ \ldots)$

Probability of Relevance

- What is $P_{true}(R| \dots)$?
  - User submits a query and he either “likes” the document or not
  - Depend on the user’s opinion
    - User decides relevant or not, then what’s the “probability” thing?

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  - Relevance depends on factors that algorithm cannot observe

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- Different users may disagree on relevance of the same doc
  - Even similar users, doing the same task, in the same context
  - $P_{true}(R|Q,D)$:
    - Proportion of all unseen users / contexts / tasks for which D would have been judged relevant to Q

PRP as Best Possible Ranking

- Let $D_i = \{Document_i, Query, User, Task, Context, \ldots\}$
- Rank documents by $p_i = p(R_i = 1 | D_i)$

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- PRP gives highest:
  - Expected precision at rank r
    - $precision_r = \frac{\#rel_r}{r} = \frac{1}{r} \sum_{i=1}^{r} \begin{cases} 1 \ldots D_i \in rel \\ 0 \ldots D_i \notin rel \end{cases}$

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    - $E[precision_r] = \frac{E[rel_r]}{r} = \frac{1}{r}\sum_{i=1}^{r}\begin{cases}1 \ldots w.p.\,p(R = 1|D_i)\\ 0 \ldots w.p.\,p(R = 0|D_i)\end{cases} = \frac{1}{r}\sum_{i=1}^{r} p_i$

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How Do We Estimate P(relevant|query,document)?

- Do not know exact probabilities, have to use estimates
- Binary Independence Retrieval (BIR) is the simplest model

Binary Independence Model

Robertson and Sparck Jones 1976

- Assumption A0:
  - Relevance of D doesn’t depend on any other document
    - Made by almost every retrieval model (exception: cluster-based)
    - Really, it’s bad to keep on returning **duplicates**

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  - $P(R|D,q) \cong \frac{P(R|D,q)}{P(NR|D,q)}$
  - x/(1-x) : monotonic function

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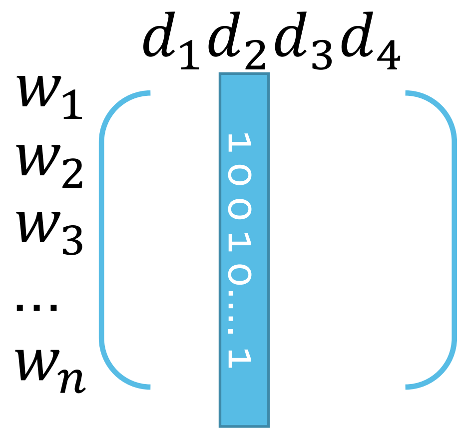
  - x/(1-x) : monotonic function
- Why Bayes? want a generative model
  - P(observation|class) sometimes easier with limited data
    - Bernoulli Naïve Bayes model (cf. text categorization!)
  - What is the purpose of P(R) and P(NR)?

Assumptions

- We want $P(D|R,q)$ and $P(D|NR,q)$
- **(Binary)** A1: $D = \{D_w\}$ one random variable for every word w
  - Bernoulli values (0,1) – word either present or absent in a document
- **(Independence)** A2: D_w are mutually independent given R
  - Terms occur in documents independently
- Allows us to write:
  - $P(R|D,q) \cong \frac{P(D|R,q)}{P(D|NR,q)} = \frac{\prod_w P(D_w|R,q)}{\prod_w P(D_w|NR,q)}$



Assumptions

- Define $p_w = P(D_w = 1|R, q)$ and $r_w = P(D_w = 1|NR, q)$

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- Define $p_w = P(D_w = 1|R,q)\ and\ r_w = P(D_w = 1|NR,q)$
- A3: $P\left(\vec{0}\middle|R\right) = P\left(\vec{0}\middle|NR\right)$
  - $P(R|D) \cong \frac{\prod_w P(D_w|R,q)}{\prod_w P(D_w|NR,q)}$ = $\prod_w \begin{cases} p_w \ldots w \in D \\ 1 - p_w \ldots w \notin D \end{cases}$ / $\prod_w \begin{cases} r_w \ldots w \in D \\ 1 - r_w \ldots w \notin D \end{cases}$ =

  = $\prod_{w \in D}(\frac{p_w}{r_w}) \prod_{w \notin D}(\frac{1 - p_w}{1 - r_w})$ -> $\prod_{w \in D}(\frac{p_w}{r_w}) \prod_{w \notin D}(\frac{1 - p_w}{1 - r_w})$ / $\underbrace{\prod_w(\frac{1 - p_w}{1 - r_w})}_{\frac{P(\vec{0}|R)}{P(\vec{0}|NR)} = 1}$ ->

- $\boldsymbol{P(R|D) \cong \prod_{w \in D} \frac{p_w(1 - r_w)}{r_w(1 - p_w)}}$

Estimation **With** Relevant Examples

$\boldsymbol{P(R|D) \cong \prod_{w \in D} \frac{p_w(1-r_w)}{r_w(1-p_w)}}$

- Suppose we have (partial) relevance judgments
  - $D_r \ldots relevant, D_{nr} \ldots non\ relevant\ documents\ marked$
  - Word w observed in $D_r(w), D_{nr}(w)$ docs
  - $P(w)$ =% of docs that contain at least one occurrence of w
    - Includes crude smoothing: avoids zeros, reduces variance

$$P_w = \frac{D_r(w)+0.5}{D_r+1.0}, r_w = \frac{D_{nr}(w)+0.5}{D_{nr}+1.0}$$

- What if we don't have relevance information?

Estimation **With** Relevant Examples

$P(R|D) \cong \prod_{w \in D} \frac{p_w(1-r_w)}{r_w(1-p_w)}$

- Relevant docs: $X_1 = \text{"a b c b d"}, X_2 = \text{"a b e f b"}$
- Non-relevant: $X_3 = \text{"b g c d"}, X_4 = \text{"b d e"}, X_5 = \text{"a b e g"}$
- Word: a b c d e f g h

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- Word:

Word:	a	b	c	d	e	f	g	h	
$D_r(w)$:	**2**	**2**	**1**	**1**	**1**	**1**	**0**	**0**	$D_r = 2$
$D_{nr}(w)$:	**1**	**3**	**1**	**2**	**2**	**0**	**2**	**0**	$D_{nr} = 3$
p_w:	$^{2.5}/_{3}$	$^{2.5}/_{3}$	$^{1.5}/_{3}$	$^{1.5}/_{3}$	$^{1.5}/_{3}$	$^{1.5}/_{3}$	$^{0.5}/_{3}$	$^{0.5}/_{3}$	
r_w:	$^{1.5}/_{4}$	$^{3.5}/_{4}$	$^{1.5}/_{4}$	$^{2.5}/_{4}$	$^{2.5}/_{4}$	$^{0.5}/_{4}$	$^{2.5}/_{4}$	$^{0.5}/_{4}$	

- New document $D_6 = \text{"}b\ g\ h\text{"}$

Estimation **With** Relevant Examples

$P(R|D) \cong \prod_{w \in D} \frac{p_w(1-r_w)}{r_w(1-p_w)}$

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- Word:

Word:	a	b	c	d	e	f	g	h	
$D_r(w)$:	2	2	1	1	1	1	0	0	$D_r = 2$
$D_{nr}(w)$:	1	3	1	2	2	0	2	0	$D_{nr} = 3$
p_w:	$^{2.5}/_3$	$^{2.5}/_3$	$^{1.5}/_3$	$^{1.5}/_3$	$^{1.5}/_3$	$^{1.5}/_3$	$^{0.5}/_3$	$^{0.5}/_3$	
r_w:	$^{1.5}/_4$	$^{3.5}/_4$	$^{1.5}/_4$	$^{2.5}/_4$	$^{2.5}/_4$	$^{0.5}/_4$	$^{2.5}/_4$	$^{0.5}/_4$	

- New document $D_6 =$ "$b\ g\ h$“
- $P(R|D_6) \cong \prod_{w \in D_6} \frac{P_w(1-r_w)}{r_w(1-p_w)}$

Estimation With Relevant Examples

$$P(R|D) \cong \prod_{w \in D} \frac{p_w(1-r_w)}{r_w(1-p_w)}$$

- Relevant docs: D_1 = "a b c b d", D_2 = "a b e f b“
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- Word:

Word:	a	b	c	d	e	f	g	h	
$D_r(w)$:	2	2	1	1	1	1	0	0	$D_r = 2$
$D_{nr}(w)$:	1	3	1	2	2	0	2	0	$D_{nr} = 3$
p_w:	$2.5/3$	$2.5/3$	$1.5/3$	$1.5/3$	$1.5/3$	$1.5/3$	$0.5/3$	$0.5/3$	
r_w:	$1.5/4$	$3.5/4$	$1.5/4$	$2.5/4$	$2.5/4$	$0.5/4$	$2.5/4$	$0.5/4$	

- New document $D_6 = "b\ g\ h$“
- $P(R|D_6) \cong \prod_{w \in D_6} \frac{p_w(1-r_w)}{r_w(1-p_w)} = \frac{\frac{2.5}{3}(1-\frac{3.5}{4})\frac{0.5}{3}(1-\frac{2.5}{4})\frac{0.5}{3}(1-\frac{0.5}{4})}{\frac{3.5}{4}(1-\frac{2.5}{3})\frac{2.5}{4}(1-\frac{0.5}{3})\frac{0.5}{4}(1-\frac{0.5}{3})}$

Estimation **Without** Relevant Examples

- Assumption A4:
  - If the word is not in the query, it is equally likely to occur in relevant and non-relevant populations
  - Practical reason: restrict product to query – document overlap
  - $p_w = r_w \ldots w \notin Q$
- Assumption A5:
  - On average, a query word will occur in half the relevant documents
  - Practical reason: p_w and (1- p_w) cancel out
  - $p_w = 0.5 \ldots w \in Q$
- Assumption A6:
  - Non-relevant set approximated by collection as a whole
    - Most documents are non-relevant
    - $r_w \approx \frac{N_w}{N} -> smoothing: \frac{N_w+0.5}{N+1}$

$$P(R|D,q) \cong \prod_{w \in D} \frac{P_w(1-r_w)}{r_w(1-p_w)} = \prod_{w \in D \cap Q} \frac{P_w(1-r_w)}{r_w(1-p_w)} = \prod_{w \in D \cap Q} \frac{1-r_w}{r_w} = \prod_{w \in D \cap Q} \overbrace{\frac{N-N_w+0.5}{N_w+0.5}}^{\text{IDF}}$$

Estimation **Without** Relevant Examples

- Documents: D1 = “a b c b d “, D2 = “b e f b”, D3 = “b g c d”, D4 = “b d e”,
  D5 = “a b e g”,
  D6 = “ b g h”

Word:	a	b	c	d	e	f	g	h	
N(W)	**2**	**6**	**2**	**3**	**3**	**1**	**3**	**1**	N=6
$\frac{N-N_w+0.5}{N_w+0.5}$	$\frac{4.5}{2.5}$	$\frac{0.5}{6.5}$	$\frac{4.5}{2.5}$	$\frac{3.5}{3.5}$	$\frac{3.5}{3.5}$	$\frac{5.5}{1.5}$	$\frac{3.5}{3.5}$	$\frac{5.5}{1.5}$	

- Query: Q = “a c h”

- $P(R\ |D_1, Q) \cong \prod_{w=Q\cap D_1} \frac{N-N_w+0.5}{N_w+0.5} = \frac{\mathbf{4.5}}{\mathbf{2.5}} * \frac{\mathbf{4.5}}{\mathbf{2.5}}$

- $P(R\ |D_2, Q) \cong 1$

- $P(R\ |D_3, Q) \cong \frac{\mathbf{4.5}}{\mathbf{2.5}}$

Classical Probabilistic Model: BIR (Binary Independence Retrieval)

- PRP: best possible ranking

  $P(R|D,q) \cong \prod_{w \in D} \frac{P_w(1-r_w)}{r_w(1-p_w)} = \prod_{w \in D \cap Q} \frac{N-N_w+0.5}{N_w+0.5} \cong \sum_{w \in D \cap Q} log \frac{N-N_w+0.5}{N_w+0.5}$

- 7 assumptions (A0-A6)

- How we can improve the model?
  - Relax particularly bad assumptions

Good and Bad News

- Standard Vector Space Model
  - Empirical for the most part
  - Few properties provable
- Probabilistic Model advantages
  - Theoretically justified optimal ranking scheme
  - Notion of relevance is directly addressed
  - Explicit, elegant model of relevance (if observable)
- Disadvantages
  - Not using term frequencies
  - Independence of terms
  - Very problematic if relevance not observable